

Sunny and shadow atmospheric fluctuations - Experiment description

Group 15

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1 Introduction

In this experiment, the aim was to determine the effect of the sun on atmospheric conditions and to see how accurately the Arduino Uno and P/T/RH sensor set-up measures these parameters. The experiment was performed in the Botanical Gardens at Utrecht University. The parameters measured for this aim were pressure(Pa), temperature(C) and humidity(%) . The experimental values are then compared to the weather report for that day.

2 Materials and Methods

2.1 Experimental set-up

Components used in the set up: Breadboard, Arduino Uno, Cables, P/T/RH sensor(BME280), SD card, SD card reader and External battery (powerbank).

The set-up is connected in the following way:

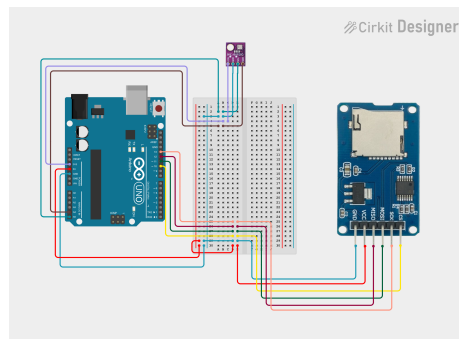


Figure 1: P,T,RH set-up scheme

SD card

- **CS** : Digital Pin 10
- **SCK** : Digital Pin 13
- **MOSI**: Digital Pin 11
- **MISO**: Digital Pin 12
- **VCC**: 5V Power Supply
- **GND**: Ground (GND)

P/T/RH(BMP280) sensor

- **VIN**: 3.3V Power Supply
- **GND**: Ground (GND)
- **SCL**: Analog Pin A5
- **SDA**: Analog Pin A4

To better view the scheme you can download 1 from: [Experiment repository](#), it has the name 'P,T,RH set-up scheme'.

The Arduino is plugged to a computer from which it can be controlled using a code. The code is made with Arduino IDE.

2.2 Collecting data

The data was acquired using the aforementioned experimental set-up and the scripts: "Erase SD card", "Recording data on P,T,RH sensor" and "Extract data from csv file", downloadable from [Experiment repository](#). The scripts should be run in the following order:

1. WITHOUT THE P/T/RH sensor(BME280) connected to breadboard, check that the SD card is working properly by running "Extract data from csv file". If all is fine "Initializing SD card... SUCCESS!" will be printed, if not "Initializing SD card... FAILED!" will be printed.
2. If SD card is working properly, erase existing data from SD card by running the "Erase SD card" script.
3. CONNECT THE P/T/RH sensor (BME280)(as shown in Experiment set-up) and run "Recording data on P,T,RH sensor". After this you should have a "data.csv" file on the SD card.

4. In order to access the "data.csv" run the script "Extract data from csv file".

2.3 Study site

The experiment was performed outside, on a metal table with benches in the Utrecht University Botanical Gardens at the exact location: 52.087706,5.169944. The sunny portion was performed at time 16:05-16:10 and the shadown portion was performed at time 16:17-16:22 on the date 24.09.2026. The location did not change between the two portions, but to create shadow a backpack was propped on the table near the Experimental set-up.

3 Results

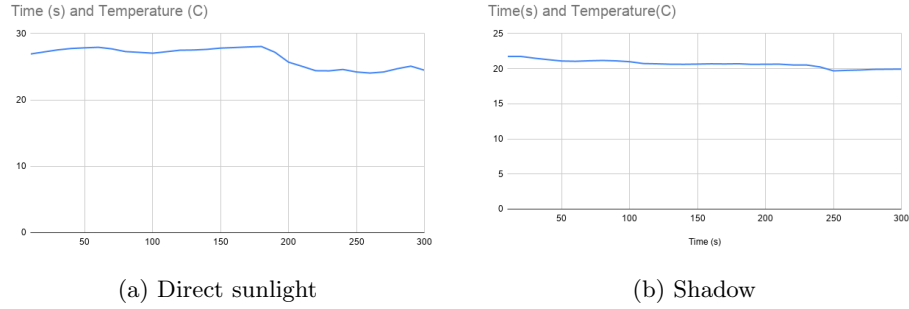


Figure 2: Temperature(C) against time(s) under different atmospheric conditions: (a) direct sunlight, (b) shadow.

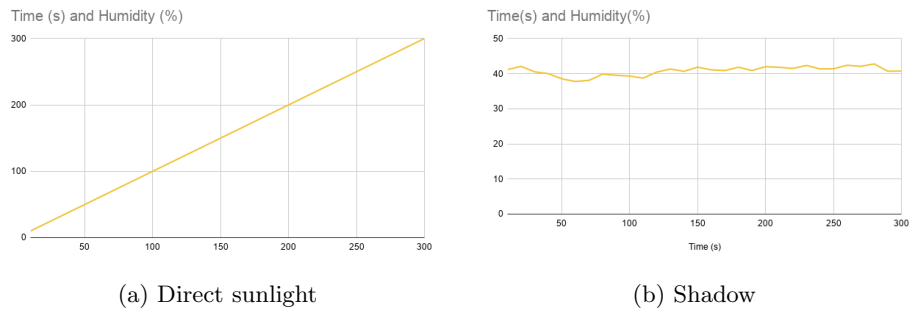
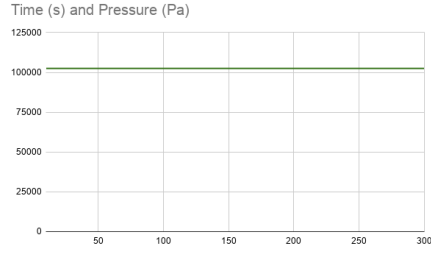
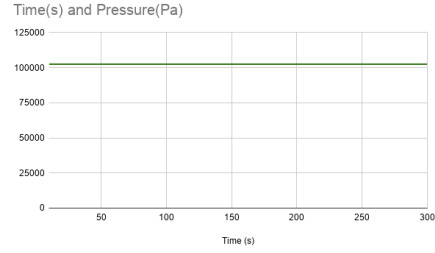


Figure 3: Humidity(%) against time(s) under different atmospheric conditions: (a) direct sunlight, (b) shadow.



(a) Direct sunlight



(b) Shadow

Figure 4: Pressure(Pa) against time(s) under different atmospheric conditions: (a) direct sunlight, (b) shadow.

4 Discussion

Table 1: Measured vs. Official Meteorological Conditions (Utrecht, 24 Sep 2026, 16:00)

Parameter	Shadow	Sunny	Station (Utrecht)
Temperature ($^{\circ}\text{C}$)	20.68	26.39	21.50
Pressure (Pa)	102486.72	102488.53	102480.00
Humidity (%)	40.79	29.90	48.00

Average values of the measured atmospheric conditions were compared to the meteorological conditions reported for 52.087706,5.169944 at 16:00 on 24.09.2026.

5 Conclusion

We hope you have fun reproducing our experiment.